

Math Fundamentals

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Part I

Introduction

I.1 Negative Numbers

When we learn how to do math for the first time, we learn addition by counting up. We end up learning how to undo addition, known as subtraction, which we can think of as counting left instead of right. What happens when you subtract a big number from a small one? The result is a negative number, but this might not feel intuitive. I like to do it by taking off small pieces at a time, which we can visualize here:

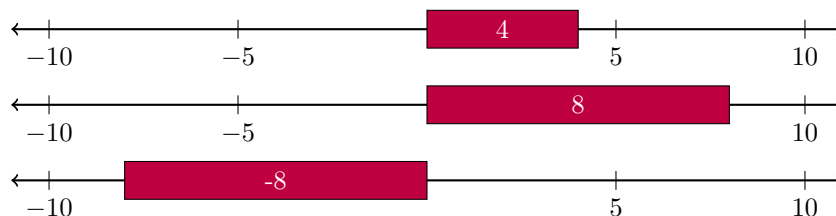
$$\begin{aligned} 6 - 17 \\ 6 - 6 - 11 & \quad \text{(Break up the 17)} \\ 0 - 11 \end{aligned}$$

So, $6 - 17 = -11$. This is the same as if we did $17 - 6$ first and then made it negative (since we took big from small). What about subtracting from a negative number? This can be thought of as adding them together and then making the result negative. After all, starting on the left and then counting to the left means you end up farther to the left.

Multiplication and division with negative numbers is almost the same as with positive numbers (positive numbers have no rules to worry about.) The only difference is that multiplying two negatives makes it positive while multiplying a positive and a negative makes it negative. It's important to know that the rule extends to multiplying a bunch of stuff together. No matter how many things you are multiplying, if you multiply an odd amount of negative numbers, then the result will be negative; if you multiply an even amount, then the result is positive.

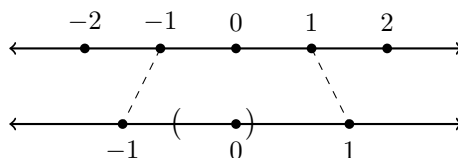
Remember: Matching multiplies positive, mismatching multiplies negative. Negative numbers are just positive numbers that have been multiplied by -1 .

We can think of addition as moving right, subtraction as moving left, multiplication as stretching, and division as shrinking. How do negatives fit in? Think of $4 \cdot (-2)$ and rewrite it as $4(-1)(2)$, then multiply 4 by 2, then finally by -1 :



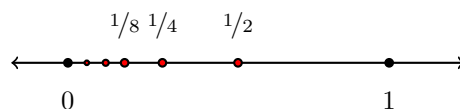
I.2 Fractions

Most of the numbers that we deal with day-to-day are integers — whole numbers, including negatives. But there are more types than just the ones you can count on your fingers. If we look at a number line and zoom in,



we can see that there are gaps in between them. Each of these gaps, from one to the next, is of length 1. If we look through a window that is just that wide, we can see pretty quickly that we will be able to see one number at most.

In the same way that repeated counting is addition, repeated addition is what we call *multiplication*. Since we can undo addition, we can undo multiplication with *division*; by subtracting a big number from a small one, we create negative numbers. What happens if we take one number and divide it by another that doesn't fit evenly? We end up with fractions, and with these, we can pretty clearly see that there are no more gaps in the number line:



Since we created them by using operations with numbers, fractions are numbers themselves (not a controversial statement), which in turn means that we can do all of the operations that we're familiar with — adding, subtracting, multiplying, and dividing. Simplifying is the most important thing about them, so we'll start with that. Addition and subtraction may require some multiplication from us, so we'll multiply and divide fractions first.

I.2.1 Simplifying

Unlike whole numbers, it's possible to write a fraction that is actually the same as another one! We need to know how to simplify fractions before we do anything further, so let's look at an example.

$$\frac{42}{76}$$

← Numerator
← Denominator

We need to break down the numerator and the denominator into their factors, which are the smallest numbers that multiply to make them. If you're not sure what all of the factors are, it isn't difficult to find them. All we need to do is to try and divide by small numbers and move on to the next when we can't anymore. Starting with 2, we do $42 \div 2 = 21$ and find that we can't divide by 2 anymore. Now we move on to 3: $21 \div 3 = 7$. It turns out that nothing will divide 7, so we are at the end. Take everything that we divided by, as well as the end result, and put them together with multiplication between them: we have $42 = 2 \cdot 3 \cdot 7$.

Let's do the same for 76. Starting with 2, we find $76 \div 2 = 38$. Divide by 2 again (we'll do this until we can't!) to get $38 \div 2 = 19$. We can't go further, so we have $76 = 2 \cdot 2 \cdot 19$. Now we can look at the fraction and rewrite it:

$$\frac{2 \cdot 3 \cdot 7}{2 \cdot 2 \cdot 19}$$

What happens next is division; specifically, the 2 on top is divided by the 2 on the bottom. You may notice that we have an extra 2 in the denominator — that one goes unused because numbers divide in a one-for-one way, which we call canceling.

$$\frac{\cancel{2} \cdot 3 \cdot 7}{\cancel{2} \cdot 2 \cdot 19} = \frac{3 \cdot 7}{2 \cdot 19} = \frac{21}{38}$$

If it's clear what factors will cancel, you don't need to break each of them down all the way. Just be sure that you're not making any mistakes; if you're not entirely certain, it's better to do more work and validate that you were correct than to learn that you were wrong!

$$\frac{36}{20} = \frac{\cancel{4} \cdot 9}{\cancel{4} \cdot 5} = \frac{9}{5}$$

Remember: Division is the same as multiplying by the **reciprocal** and has the upside of not having confusing notation. A reciprocal is a number means to “flip” the fraction; 4 is $\frac{1}{4}$, 8 is $\frac{1}{8}$, $\frac{5}{1}$ is 5, $\frac{4}{7}$ is $\frac{7}{4}$, and so on. So, it’s best to treat division like this:

$$4 \div 8 = \frac{4}{8} = \frac{1}{2}$$

I.2.2 Multiplication

Multiplying fractions is extremely simple: you don’t need to do anything besides multiply straight across. Don’t forget that you need to simplify your answer! Jumping into an example, remember that multiplication means that we can rearrange this however we want:

$$\frac{20}{9} \cdot \frac{15}{4} = \frac{20 \cdot 15}{9 \cdot 4} = 20 \cdot 15 \cdot \frac{1}{9 \cdot 4} = \frac{20}{1} \cdot \frac{15}{1} \cdot \frac{1}{9 \cdot 4}$$

I recommend that you rearrange in a way that makes it clear what you should do. There are two ways you can do this; you can decide to multiply first and then simplify, or you can simplify first and then multiply. Multiplying first is easy to do but is a pain since you may end up with larger numbers, so we’ll simplify first instead. We’ll put everything into one fraction as well:

$$\begin{aligned} \frac{20}{9} \cdot \frac{15}{4} &= \frac{4 \cdot 5 \cdot 3 \cdot 5}{3 \cdot 3 \cdot 4} \\ &= \frac{4}{4} \cdot \frac{3}{3} \cdot \frac{5 \cdot 5}{3} \\ &= \frac{25}{3} \end{aligned} \quad \text{(Rearranged for visual aid)}$$

I.2.2.1 Division

Dividing fractions is nearly the same as multiplying them, with just one additional step at the beginning: whatever is doing the dividing is flipped, then multiplied. This is called “keep, change, flip.”

$$\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \cdot \frac{2}{1} = \frac{3}{2}$$

Sometimes it may be presented in a bit of a confusing way, with a fraction over a fraction, but the process doesn’t change. Just flip the fraction on the bottom and multiply:

$$\frac{\frac{8}{9}}{\frac{6}{5}} = \frac{8}{9} \cdot \frac{5}{6} = \frac{20}{27}$$

I.2.3 Addition

Fractions can’t be added unless they have the same denominator; if they do, then we just add (or subtract) the numerator. For example, $\frac{3}{7} + \frac{2}{7} = \frac{5}{7}$. With different denominators, we have a simple method: cross-multiplying. It looks like this:

$$\frac{2}{3} + \frac{7}{6} = \frac{6 \cdot 2 + 7 \cdot 3}{6 \cdot 3} = \frac{33}{18} = \frac{11}{6}$$

While it’s good enough, it’s important to dispel some of the magic. What are we actually doing when we do this? You should know that multiplying anything by 1 is itself, so doing this doesn’t change anything:

$$\frac{2}{3} + \frac{7}{6} = \frac{6}{6} \cdot \frac{2}{3} + \frac{7}{6} \cdot \frac{3}{3}$$

In other words, we multiply each term by the other's denominator. This is what's actually happening when we cross multiply, and we're just ignoring this important middle step. Why should we care? Notice that the denominators of $\frac{2}{3}$ and $\frac{7}{6}$ have a factor in common: namely, $6 = 2 \cdot 3$. With just cross multiplying, we had to cancel in order to simplify our answer. But if we decide to just multiply the terms by the factors that they're missing, we won't need to simplify in the end:

$$\begin{aligned}\frac{2}{3} + \frac{7}{2 \cdot 3} &= \frac{2}{2} \cdot \frac{2}{3} + \frac{7}{6} \\ &= \frac{4}{6} + \frac{7}{6} \\ &= \frac{11}{6}\end{aligned}$$

I.3 Advanced Operations

I.3.1 Parenthesis

Operations contained in a set of parenthesis tell us “do this first”. If you're using a calculator, be sure to not make the mistake of misplacing parenthesis or not using them when you should. It's possible to have nested parenthesis as well; in this case, you want to work on the innermost set first. It would be unusual to have something that looks like a Russian nesting doll, so be sure you understand what you're trying to write.

It is extremely important to know that any time there are two objects next to one another, without a sign between them, they are being multiplied together. $4x$ means “four times x ”, and $5(2 - 6)$ means “five times the difference of two and six”. Personally, I wouldn't write this as $5 \cdot (2 - 6)$ unless I wanted to highlight that I was making some sort of change.

I strongly recommend that you use parenthesis whenever you make a substitution or want to keep track of the changes that you are making from one step to the next. It is **required** that you use parenthesis when you are multiplying negative numbers. After all, it's easy to understand $-2 \cdot 4$, but writing $4 \cdot -2$ is confusing if not utterly meaningless.

I.3.2 Exponents

In the same way that multiplication is repeated addition, repeated multiplication is represented by exponents. Simply put, 6^3 means multiplying 6 by itself 3 times — $6 \cdot 6 \cdot 6$. There are a few different ways to read this:

- six to the third power (or just “to the third”)
- six to the three
- six cubed

On the other hand, for example, 5^2 could be read as “5 squared”. The reason we say “squared” for a power of two is because a square has two dimensions, height and width; “cubed” means a power of three because cubes add another dimension called depth (or length, but it doesn't matter). It should be obvious that exponents get very big very quickly. Let's 10 and 5 in three ways:

$$10 + 5 = 15, \quad 10 \cdot 5 = 50, \quad 10^5 = 100,000$$

and it should *also* be obvious that order matters:

$$5^{10} = 9,765,625$$

As a consequence of the definition, even exponents will always have even results. But odd exponents may be negative:

$$\begin{aligned} (-2)^4 &= \boxed{(-2) \cdot (-2)} \cdot \boxed{(-2) \cdot (-2)} & (-2)^3 &= \boxed{(-2) \cdot (-2)} \cdot (-2) \\ &= 4 \cdot 4 & &= 4 \cdot (-2) \\ &= 16 & &= -8 \end{aligned}$$

It's worth talking about variables now. As expected, $x \cdot x$ means x^2 ; if you see $x^2 \cdot x$, know that this means multiplying x by itself *three* times in total and is equal to x^3 . Here are some more examples:

$$y^2 \cdot y^3 = y^5 \qquad (x^5 z) \cdot (z^2 y^9) = x^5 y^9 z^3 \qquad (xz^3)^2 = x^2 z^6$$

If you can identify the pattern, you understand the rule for exponents. Here's one more example that's fairly complicated; if you understand the rule, then this should make sense too:

$$\frac{x^2 y^3 z^{x+2}}{x^5 y^2 z} = \frac{y z^{x+1}}{x^3}$$

In the same way that subtraction and division undo addition and multiplication, we can undo exponentiation with *roots*. We can have square roots, cube roots, seventeenth roots, and so on:

$$\sqrt{25} = 5 \qquad \sqrt[3]{-64} = -4$$

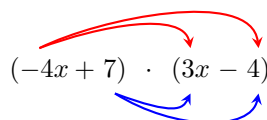
To wrap things up, we need to make an important clarification. A function has exactly one output per input and this means that $f(x) = \sqrt{x}$ would not be a function if we allowed $\sqrt{4} = \pm 2$ (shorthand to say both positive and negative 2). To keep things straight, the definition the square root of a number is **always** positive. However, when we solve an equation like $x^2 = 4$, it can be the true that $x = 2$ or $x = -2$. Only in the case of solving for a variable can we write $x = \pm\sqrt{4}$

I.3.3 Distribution

If you want to add two things together and multiply them, how do you do it? The relationship between addition and multiplication gives us what's called the **distributive property**. When distributing, you will multiply *each to each*: each term on the left will multiply each term on the right, and everything is recombined with addition. To solve $-3(2x - 10x)$, we need to multiply $2x$ and -10 by -3 , then add the results together:

$$\begin{aligned} -3(2x - 10) &= (-3)(2x) + (-3)(-10) \\ &= -6x + 30 \end{aligned}$$

For a bigger example, let's multiply expressions that contain two terms. We'll do this in two different ways; right now, we'll use *each to each*. First is a diagram of how things multiply:



Now let's use some colors to keep track of our work:

$$\begin{aligned} ((-4x) + 7)(3x - 4) &= \boxed{-4x(3x)} + \boxed{-4x(-4)} + \boxed{7(3x)} + \boxed{7(-4)} \\ &= \boxed{-12x^2 + 16x} + \boxed{21x - 28} \\ &= -12x^2 + 37x - 28 \end{aligned}$$

On the other hand, using the power of addition, we can “break up” one of the terms that we are multiplying. Let’s get crazy and use the term on the right instead of the left, since order doesn’t matter when you multiply:

$$\begin{aligned} (-4x + 7)(\boxed{3x}\boxed{-4}) &= \boxed{3x(-4x + 7)} + (\boxed{-4(-4x + 7)}) \\ &= \boxed{-12x^2 + 21x} \boxed{+16x - 28} \\ &= -12x^2 + 37x - 28 \end{aligned}$$

Notice that our first step could be written as $3x(-4x + 7) - 4(-4x + 7)$. Writing it like this is perfectly fine, but there’s a chance that you might forget about the fact that you’re multiplying by a negative number. It’s also perfectly fine to distribute the 4 and keep the minus, like this,

$$-12x^2 + 21x - (-16x + 28)$$

but you’ll have to remember to respect the parenthesis and later multiply by the -1 (which is to say, flip the signs inside). It’s very easy to forget about this! To avoid any trouble in these cases, it’s best to draw parenthesis around everything and use a plus sign instead. Below is a comparison so that you can see the difference.

$$\begin{array}{ll} 3x(-4x + 7) - 4(-4x + 7) & \text{(Left alone)} \\ 3x(-4x + 7) + (-4(-4x + 7)) & \text{(My recommendation)} \end{array}$$

It may also help to recall that a negative number is the positive version multiplied by -1 , so you can even write it like this if it helps:

$$3x(-4x + 7) + (-1)(4)(-4x + 7)$$

Warning: Exponents **do not** distribute over addition. Remember that exponents represent multiplying by itself! The statement below is *not* true, and you should check it for yourself by plugging in different numbers for x :

$$(x + 2)^2 = x^2 + 2^2$$

However, exponents *do* distribute over multiplication, but this is for another time.

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Part II

Modeling With Functions

Part III

Linear Functions

Part IV

Equations & Inequalities in One Variable

IV.1 Equations and Expressions

In math, not everything gets the explanation that it deserves. For example, the words “expression” and “equation” appear in similar contexts, but there have to be *some* differences — otherwise, why would we bother using different words? — which come down to purpose. An **equation** makes a statement about equality between things, while an **expression** simply expresses an operation. It wouldn’t be wrong to think of an expression as being *almost* an equation. If we write $5x + y - 3$ we have an expression, but if we write $5x + y - 3 = 0$, we now have an equation! The key difference for you is that equations can be *solved*, while expressions can only be *simplified*.

Another thing that often goes unexplained is the usage of $f(x)$ versus y . To be short, there is **no difference** between writing $f(x) = 4x + 5$ and $y = 4x + 5$. The reason that we would use one or the other depends on purpose: it’s far easier to say $f(x) =$ or $g(x) =$ when there are multiple equations to consider, while using $y =$ might be better when you want to think about graphing and finding points. $f(x)$ is read as “f of x”, short for “f is a function of the variable x”, so you can just as easily say f when referring to the function. From here on, we’ll try to use $f(x)$ as often as possible because of how useful it is. For the rest of this section, we’ll use $f(x) = -4x + 5$ as an example.

IV.1.1 Types of Functions

Many times, we use the word “equation” to describe a function. While the equations you need to be able to recognize don’t change, the they are classified can differ depending on the perspective. I put the ones that you’ll need to care about into two categories:

1. **Constant:** No variables, just a number
2. **Polynomial:** Constants and variables whose degrees are positive whole numbers
 - (a) **Linear:** Highest degree of the variable is 1 (the variable has no exponent written)
 - (b) **Quadratic:** Highest degree of the variable is 2
 - (c) **Cubic:** Highest degree of the variable is 3
3. **Exponential:** The variable is in the exponent

IV.1.2 Structure

Here are some definitions to identify the structure of an equation or expression.

1. **Terms:** The parts that are separated by addition or subtraction.
 - $f(x)$ has two terms, $-4x$ and 5
2. **Constant:** A term that is not a variable, that never changes, and is constantly the same; a number by itself.
 - $f(x) = -4x + \textcircled{5}$
3. **Variable:** A term that contains a quantity that can change, symbolized by a letter.
 - $f(x) = -4\textcircled{x} + 5$
4. **Coefficient:** The part of a term that multiplies a variable.
 - $f(x) = \textcircled{-4}x + 5$

IV.1.3 Solving

When we write $f(-2)$, what we want is to evaluate the equation f at -2 — this is a fancy way of saying to plug -2 in for x wherever you see it. It's generally good practice to write parenthesis whenever to plug something in or make a substitution so that you can keep track of the signs, since it's easy to lose track of a negative sign when you multiply!

$$\begin{aligned} f(4) &= -4(-2) + 5 && \text{(Substitute -2 for } x\text{)} \\ &= 8 + 5 && \text{(Multiply -4 and -2)} \\ &= 13 \end{aligned}$$

IV.1.4 Rearranging

When we are given an equation, sometimes we are instructed to solve it in terms of a certain variable. In this sense, it is our task to *rearrange* the equation by undoing the operations that are present around that variable; we want to get it all by itself by using the opposite operations: addition/subtraction, multiplication/division, and exponents/roots. For example, we have Newton's formula for the force of gravity between two objects; their distance is r , their masses are m_1 and m_2 , and G is the gravitational constant.

$$F_g = G \frac{m_1 m_2}{r^2}$$

You could read this as “F g equals capital G times m 1 times m 2 divided by r squared”. Here's how we solve for r :

$$\begin{aligned} F_g &= G \frac{m_1 m_2}{r^2} && \text{(Multiply both sides by } r^2\text{)} \\ \textcircled{F_g} r^2 &= G m_1 m_2 && \text{(Divide by } F_g\text{)} \\ r^{\textcircled{2}} &= G \frac{m_1 m_2}{F_g} && \text{(Take the square root of } r\text{)} \\ r &= \sqrt{G \frac{m_1 m_2}{F_g}} \end{aligned}$$

However, you have to remember to keep both sides of the equation the same. They are in balance to start with; don't unbalance it by making changes to one side and not both! Here's a quick guide to doing this:

1. Identify the variable that you're solving for. How many terms are on the side of the equation, and how many have the variable?
2. **Goal:** Reduce the number of terms on every side.
 - (a) Combine like terms and factor out the variable, if possible.
3. **Goal:** Isolate the term containing the variable
 - (a) Add or subtract the other terms on the side of the variable to cancel them; do it on the other side too.
 - (b) Multiply or divide to remove the coefficient attached to the variable; do it on the other side too.

Note. When you divide or multiply to isolate a variable, you must match the sign of the coefficient. If you have $-4x = 20$, you need to divide both sides by -4 and not just 4. If you're left with a negative sign in front of the variable, just multiply everything by -1 .

IV.1.4.1 Inequalities

An inequality represented by a sign that indicates if something is more or less. There are also signs that allow for equality: $5 < 5$ is not true because 5 is not less than itself, but $5 \geq 5$ *is* true because 5 is certainly greater than or equal to itself.

Inequalities and equations are approached in the same way. In fact, they are nearly the same except that it is almost a guarantee that many different numbers will make an inequality true. The primary difference in solving them is with multiplication or division by negative numbers. In the same way that multiplying a negative is like turning around, the inequality will “flip” as well. For example,

$$\begin{aligned}8 - 3x &\leq 26 \\ -3x &\leq 18 \\ x &\geq 6\end{aligned}$$

So all numbers greater than or equal to x make the inequality true.

Remember: “Greater than” opens to the left while “less than” opens to the right; $>$ is greater and $<$ is less.

IV.2 Polynomials

IV.2.1 Arithmetic

Arithmetic with polynomials has been covered already! You know how to combine like terms, how to multiply, how to handle exponents, and how to distribute — just make sure you can handle variables and you’re good to go!